Metrological Domain Profiling IV: Dual Systems, Redundant Channels, and the Imperial Data Network of the Inca Khipu

Adrian Sharman — *SoulDriver Research (souldriver.com.au)*

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Abstract

The knotted-cord recording devices (khipus) of the Inca Empire (~1400–1532 CE) encode information across five simultaneous channels: knot value, cord colour, twist direction, hierarchical depth, and fibre material. While colour has long been recognised as informationally significant, no corpus-wide quantitative framework has measured how 720 distinct compound colours in the Open Khipu Repository (OKR) distribute their information load. This paper applies Shannon entropy analysis to the complete OKR colour space (39,172 pendant cords, 612 khipus) and demonstrates a three-tier information hierarchy structurally analogous to the vowel–consonant distinction in natural language alphabets. Collapsing compound colours to their primary component — the standard practice in khipu studies since the Aschers — discards 1.366 bits, fully 27% of the total colour entropy, rendering a quarter of the colour signal invisible to every previous computational study. Corpus-wide classification of all 612 khipus identifies three co-existing administrative systems: System A (Logistical/Warehouse, 154 specimens, 25.2%), characterised by cotton fibre, polychrome colour matrices, hierarchical subsidiary trees, and S-type decimal arithmetic; System B (Demographic/Governor, 28 specimens, 4.6%), characterised by camelid or human hair fibre, monochrome flat structure, L-type tactile knots, and Gaussian value distributions centred on village-scale headcounts; and System C (Categorical/Codebook, 63 specimens, 10.3%), exhibiting Zipfian power-law distributions on mixed-system hardware with a shared pan-Andean vocabulary of 11 categorical codes. Evolutionary analysis of wrapped-construction specimens and the newly identified "hair khipu" (KH0289) traces the System B recording tradition through Wari antecedents (~600–1000 CE) to a pre-Wari human-hair origin, establishing that the Inca inherited rather than invented their governance recording system. Cross-site matched-pair analysis identifies 3,512 khipu pairs sharing sequential value matches, of which 1,747 span different archaeological sites. Monte Carlo simulation confirms this exceeds chance expectation by a factor of 8.1 ($p < 0.001$, 0/1,000 null iterations). The resulting network maps a continental bidirectional administrative pipeline with three hub sites — Incahuasi, Pachacamac, and Leymebamba — spanning distances exceeding 1,000 km. Audits of the Medrano/Urton Santa Valley and Leymebamba tomb collections demonstrate that both contain exclusively System A state copies, explaining the long-standing puzzle of why most surviving khipus appear purely arithmetic: the archaeological record preserves the formatted receiver end of the postal pipeline, not the raw governor originals. All findings are validated through a three-layer "Glass Box" reproducibility framework: interactive discovery via MCP tools, algorithmic extraction into standalone Python scripts with zero dependencies, and raw SQL queries executable against the public OKR database. **Keywords:** Inca khipu, Shannon entropy, information theory, colour encoding, administrative archaeology, Wari, matched pairs, computational ar-

chaeology, Open Khipu Repository, dual systems

1. Introduction

1.1 The Spreadsheet Fallacy

For over a century, the dominant assumption in khipu studies has been that knotted cords are base-10 arithmetic ledgers — the Inca equivalent of a spreadsheet. This assumption is not unreasonable: the chronicler Garcilaso de la Vega reported in 1609 that khipukamayuqs (cord-keepers) used knots to record census data and tribute obligations, and Leland Locke demonstrated in the 1920s that knot positions encode decimal place values. The Aschers formalised this arithmetic model, and Urton and colleagues extended it to cluster summation and subsidiary hierarchies.

But a spreadsheet is a specific kind of document. It stores numbers in a rigid grid. It assumes that all values are commensurable — that every cell participates in the same arithmetic. And it treats formatting (gridlines, colours, headers) as decoration rather than data. When researchers approach a khipu as a spreadsheet, they naturally focus on whether the knot values sum correctly, whether clusters show arithmetic relationships, and whether the document "works" as a ledger. Khipus that fail these tests are classified as damaged, incomplete, or anomalous.

This paper demonstrates that the spreadsheet model applies to roughly one quarter of the surviving corpus. The remaining three quarters operate under different rules — demographic censuses that count people rather than commodities, categorical codebooks that encode administrative labels rather than quantities, and mixed-system documents that combine multiple recording traditions on a single cord. The "anomalous" khipus are not broken spreadsheets. They are different kinds of documents entirely.

1.2 The Missing Originals

The puzzle of khipu studies is not that we cannot read them, but that most surviving specimens look the same. The great curated collections — the Ascher corpus, the Harvard Khipu Database, Medrano's Santa Valley census decipherment, the Leymebamba tomb cache — are overwhelmingly cotton, polychrome, hierarchical, and arithmetic. Chroniclers described a far richer system: khipus recording laws, histories, songs, genealogies, and diplomatic correspondence. If the khipu was truly a general-purpose recording technology, where are all the non-arithmetic documents?

This paper provides a computational answer. The Inca Empire operated two parallel administrative systems on different recording media. System A — cotton, polychrome, hierarchical — was the logistics system, used in coastal warehouses to track commodities. System B — camelid fibre, monochrome, flat — was the governance system, used by highland officials to record census data, tribute assessments, and administrative categories. When a provincial governor compiled a census, he recorded it on a System B khipu — monochrome camelid cords with L-type knots encoding headcounts in a Gaussian distribution centred on village populations of 5–30. This original was then transmitted via the chasqui postal network to the capital, where state scribes transcribed the data onto a System A khipu — cotton cords with S-type knots, polychrome colour matrices, and hierarchical subsidiary trees suitable for imperial accounting.

The archaeological record preserves the *receivers*, not the *senders*. Tomb caches, museum collections, and warehouse excavations contain the endpoint of the administrative pipeline: formatted state copies on durable cotton. The camelid governance originals — working documents created in the field, carried by runners, and superseded once transcribed — were ephemeral. They survive only as rare specimens in private collections and luxury cedar boxes, far from the administrative centres where they were created.

This survival bias explains why 100 years of research has produced the impression that khipus are "just arithmetic." The arithmetic khipus are the ones that survived.

1.3 Corpus and Methods

All analyses in this paper operate on the Open Khipu Repository (OKR; Clindaniel et al., DOI: 10.5281/zenodo.18025748), accessed as a SQLite database containing 619 khipus, 54,403 cords, and 110,677 knots. Of these, 612 contain sufficient cord data for classification (7 entries have no recorded cords). The corpus was digitised by multiple independent research teams spanning four decades: the Aschers (~215 khipus), Gary Urton and the Harvard Khipu Database Project (~280), Hugo Pereyra (~57), and Jon Clindaniel (23). Provenance spans the Inca administrative geography from Playa Miller (Arica, Chile) to Leymebamba (Chachapoyas, northern Peru), with major concentrations at Pachacamac (90 khipus), Incahuasi (52), Ica/Pisco (49), and Leymebamba (22).

Initial exploration of the corpus was conducted using a Knowledge Acquisition Research Protocol (KARP) architecture, in which the OKR database was loaded into an in-memory semantic graph accessible via 25 dynamic analysis tools through a Model Context Protocol (MCP) server. This allowed rapid, conversational hypothesis testing — generating and falsifying dozens of hypotheses interactively before extracting validated algorithms into standalone scripts. The following table maps the MCP tools that directly produced findings cited in this paper:

Tool	Findings Produced	Paper Section
classify_system	F1, F11, F25, F28, F30	§3.1, §4.3
value_transitions	F2, F9, F12, F29	§3.4, §5.1
cord_sequence	F6, F23, F26	§2, §3.5
colour_transitions	F4	§2
clean_summation	F18	§3.5
scribe_fingerprint	F14	§3.3
audit_trail	F8, F19	§3.3, §7
twist_analysis	F3, F12	§3.2
metadata_query	F2, F29	§5.1
cord_notes_search	F19	§7
canuto_analysis	F25	§4.3

The methodology is described in detail in §6 ("The Glass Box Approach").

All statistical findings reported in this paper are reproduced by four standalone Python scripts requiring only the Python 3.8+ standard library and the public OKR SQLite database. No finding depends on large language model output, proprietary software, or non-public data.

2. The Colour Signal

2.1 The Shannon Entropy Hierarchy

A khipu cord encodes colour information using a compound notation system that captures barberpole patterns (two colours twisted together, denoted AB:W), striped patterns (MB-KB), and complex multi-colour constructions (W*SR:SY:0G). The OKR records these in a FULL_COLOR field containing 720 unique designations across 39,172 pendant cords. Standard practice since the Aschers has been to collapse compounds to their primary component (COLOR_CD_1), reducing the palette to 65 primaries. The information-theoretic cost of this collapse has never been measured.

Shannon entropy quantifies the average information content of a symbol system. For a palette of K colours with frequencies $f_1 \ldots f_k$ across N cords, the probability of each colour is $p(c_i) = f_i / N$, the self-information of each occurrence is $I(c_i) = -log_2(p(c_i))$ bits, and the corpus entropy is $H = -\Sigma\ p(c_i)\ log_2(p(c_i))$. The maximum entropy for a uniform distribution over K colours is $H\_max = log_2(K)$.

The 720-colour FULL_COLOR palette yields a corpus entropy of **4.995 bits** (52.6% of the theoretical maximum of 9.492 bits). The 65-colour COLOR_CD_1 palette yields **3.630 bits**. The difference — **1.366 bits** — represents **27% of the full colour entropy**. Every previous computational study of khipu colour operated on a signal with more than a quarter of its information content removed.

Three natural tiers emerge when colours are ranked by information content, forming a hierarchy structurally analogous to the vowel–consonant distinction in alphabetic writing systems.

Vowels (< 4 bits): White (W, 10,604 occurrences, 1.89 bits), Light Brown (AB, 5,338, 2.88 bits), and Medium Brown (MB, 3,838, 3.35 bits). These three colours account for 50.5% of all pendant cords. Like the high-frequency function words of a language (“the,” “of,” “and”), each individual occurrence carries minimal identifying information. Their role is structural — they define matrix positions and category boundaries, not what is being counted.

Semi-vowels (4–7 bits): Fourteen colours including B (4.52), YB (4.69), KB (5.46), NB (5.90), HB (6.37), and LK (6.49). These appear regularly but each occurrence conveys moderate information — enough to narrow the khipu’s administrative context.

Consonants (7–10 bits): Sixty colours including RL (7.29), BG (7.59), CB (8.03), and GL (9.59). Each occurrence carries substantial identifying power, analogous to the low-frequency content words that carry a sentence’s meaning.

Rare consonants (10+ bits): 643 colours appearing fewer than 40 times each, including SR (11.01 bits) and compound constructions like W*SR:SY:0G (12.45 bits). A single occurrence of SR narrows a khipu’s identity more than five consecutive white cords.

2.2 The Functional Taxonomy of Colour

Colour-value correlation analysis reveals that each entropy tier maps to a distinct administrative function, forming a three-layer encoding system:

Layer 1 — Formatting colours are defined by near-total zero-value rates. Scarlet Red (SR) achieves exactly 100% zero across the entire corpus — not a single SR cord in the OKR carries a knot value. Among Dark Brown compound variants, KB:GG:W, KB:LC, and KB:SY are likewise 100% zero. These colours function identically to gridlines, tab stops, and section dividers in a modern spreadsheet. They organise the document's structure without carrying data.

Layer 2 — Category labels (the vowels W, AB, MB) carry values spanning the full corpus range (0 to 320,535) with high standard deviations. Their function is to mark column positions — "column 1, column 2, column 3" — with the actual quantities encoded by knot values. This was confirmed by the AB-AB-AB-MB 4-gram motif, which shows 93% unique value tuples across 153 instances: the colour marks the position, not the amount.

Layer 3 — Domain identifiers carry characteristic value distributions that reflect *what is being counted*:

- **LK (Black):** 9.3% zero, mean value 21.2, max 500. The tight distribution (mean ~21, bounded at 500) is consistent with counting individuals in villages or work teams — small, bounded human populations.
- **NB (Strong Yellowish Brown):** 42.4% zero, mean value 515.7, max 10,170. The heavy tail is consistent with bulk agricultural commodities. NB was independently validated as the Munsell colour match for dried peanut shells (10YR 5/6), and clusters 100% at Incahuasi, an archaeologically documented warehouse storing peanuts.

The LK-to-NB value ratio of 25× reflects the fundamental scale difference between counting people and counting produce. A village has 20 residents; a warehouse stores 500 sacks of peanuts.

2.3 The KB Dual-Function Discovery

Dark Brown (KB, 2,002 pendant cords) presents an apparent paradox: 97% zero on some khipus but substantial values on others. This paradox resolves along two axes:

Compound suffix determines function. KB with formatting-type suffixes (KB:GG:W, KB:LC, KB:SY) is 100% zero-value — pure formatting. KB with data-type suffixes (KB:W, KB-AB) carries real data at only 31–40% zero. The compound suffix acts as a function flag, and this flag is invisible when colours are collapsed to COLOR_CD_1.

Archaeological site determines function. KB is 95.2% zero at Ica/Pisco and 97.6% at Huacho (coastal System A centres) but only 30.8% zero at Leymebamba and 18.6% at Ica town (provincial System B sites). The KB colour tracks the boundary between the two administrative systems — formatting at the coast, data in the provinces.

2.4 Site-Specific Colour Signatures

Entropy-weighted motif search using FULL_COLOR 3-grams and 4-grams identifies 266 candidate "colour words" exhibiting high entropy (>15 bits), high value variance (>60%), and geographic concentration (>50% at a single site). Six major centres show distinct colour fingerprints:

Site	Signature	Khipus	Concentration	Interpretation
Valle de Pisco	LK×4	7	98%	Governance/demographic
Incahuasi	NB×4	12	100%	Commodity (peanuts)
Nazca	G×4	3	95%	Regional identifier
Santa	RL×4	4	100%	Regional identifier
Ica/Pisco	PR×4	4	82%	Regional identifier
Pachacamac	LB:W×4	4	100%	Administrative compound

The NB-at-Incahuasi result provides independent validation: the entropy search, operating on purely statistical grounds, recovers a commodity-site association previously established through physical colour matching and archaeological excavation.

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3. Administrative Infrastructure

3.1 The Dual-System Ontology

Computational classification of all 612 khipus using a composite scoring algorithm across six to eight dimensions reveals two distinct administrative operating systems:

System A (Logistical/Warehouse) — 154 specimens (25.2%) Cotton fibre, polychrome (≥8 colours), deeply hierarchical (4–5 levels of subsidiary cords), S-type knots for positional decimal arithmetic, wide value ranges (means in hundreds, maxima in thousands). Found at coastal administrative centres: Incahuasi (garrison/warehouse), Pachacamac (oracle/data centre), Leymebamba (state archive). Cluster summation works at Incahuasi (ratio = 1.00, confirming Urton's tax formula). These are the "spreadsheet" khipus — and they account for exactly one quarter of the corpus.

System B (Demographic/Governor) — 28 specimens (4.6%) Camelid or human hair fibre, monochrome (≤2 colours), completely flat (zero subsidiaries), L-type knots for tactile ridge-counting, Gaussian value distributions centred on 7–27 (village-scale headcounts, bounded at ~50). Authenticated with canutito thread wrappings and stored in luxury containers. Found across 9 provenances spanning highland and coastal sites, recorded by 14 independent investigators. Seven specimens score above 0.50 (HIGH confidence): KH0288 (0.72), KH0280 (0.67), KH0348 (0.64), KH0346 (0.59), KH0282 (0.57), KH0384 (0.57), and KH0289 (0.50 — the human hair khipu).

The classifier assigns each khipu a composite score from −1 (strong System A) to +1 (strong System B) by averaging scores across fibre type, topology, knot syntax, colour diversity, value range, and primary cord beginning, with optional bonus dimensions for LK enrichment and canuto detection. The thresholds (≥0.3 = System B, ≤−0.3 = System A) are deliberately conservative: 430 khipus (70.3%) fall in the Mixed zone. This is a feature, not a limitation — it ensures that every System A or System B classification represents a confident assignment, and the Mixed category honestly reflects specimens that share traits of both systems.

3.2 The Redundant Channel Principle

Every khipu has at least two formatting channels available: colour and twist direction. This paper demonstrates that the Inca used these channels in complementary alternation:

System A (polychrome) uses colour for structural formatting — SR section dividers, KB cluster boundaries, HB group markers — while twist is uniform (typically 99%+ S-type). The formatting signal lives entirely in the colour channel; twist carries no structural information.

System B (monochrome) uses twist for structural formatting — Z-density phases shift from 8% to 83% across the document (KH0288), S-twist cords serve as rare section dividers, and Z-twist values are capped at 20 — while colour is uniform (98%+ single colour). The formatting signal lives entirely in the twist channel; colour carries no structural information.

The principle ensures that data and formatting never interfere on the same channel. It also means that analysing colour on a monochrome khipu, or twist on a polychrome khipu, will always yield null results — explaining many negative findings in the literature.

3.3 System B: Immutable, Manufactured, Authenticated

Three independent lines of evidence demonstrate that System B documents were treated as tamper-proof records:

Pre-planned manufacture. Cord-length coefficient of variation (CV) on System B specimens is 11.1% (KH0288) — indicating that all cords were measured and cut to uniform length before any knots were tied. System A specimens show 30%+ CV — cords were cut as needed and length varies with the magnitude of the value recorded. System B documents were manufactured to a specification; System A documents grew organically.

Zero physical edits. Only 7 cut cords exist in the entire corpus of 54,403 (0.013%). All 7 appear on System A or Mixed khipus — 6 on KH0321 (a header redaction) and 1 on KH0526. Insertions ("cord inserted between P10 and P11," "inserted through one ply") are likewise exclusive to System A. No System B specimen shows any evidence of physical modification.

Canutito authentication. System B documents carry canutito thread wrappings — small coloured threads wound around individual cords — that are 98.6% colour-independent from their parent cord. These are not decorative; they function as authentication seals, analogous to wax seals on medieval correspondence, confirming that the document has not been altered since the canutito was applied.

3.4 System C: The Categorical Codebook

Sixty-three specimens (10.3%) classified as Mixed exhibit Zipfian (power-law) value distributions — a pattern characteristic of categorical data (like word frequencies in natural language) rather than arithmetic data (like warehouse inventories). The confirmed exemplars are KH0100 (Munich, 19 different knot types) and KH0108 (Pachacamac, 23 knot clusters), which share an 11-value vocabulary covering ~47% of each specimen. The most frequent "words" are values 11 (18 occurrences), 12 (15), and 10 (14).

Monte Carlo bigram analysis (N=1,000) tests whether these values exhibit sequential grammar — whether certain value transitions are preferred or forbidden, as in natural language. Result: no significant syntax. KH0100 shows 4.8% preferred transitions versus 4.0% expected under random order; KH0108 shows 8.7% versus 7.1%. System C is not proto-writing. It is a categorical registry — a standardised Imperial codebook of administrative labels, analogous to a modern system of postal codes or ISO country codes. The values are identifiers, not quantities, and they appear on System A hardware (cotton, hierarchical) suggesting that the codebook was maintained by the state bureaucracy rather than provincial governors.

3.5 Mathematical Punctuation and Clean Math

The largest khipu in the corpus, KH0082 (Lluta Valley, 1,831 cords), uses five minority colour classes as zero-value structural markers: SR (16 cords, 100% zero, mean gap 105 cords), KB (32 cords, 97% zero), HB (8 cords, 100% zero), FB (13 cords, 100% zero), and BS (25 cords, 92% zero, appearing in pairs). These function as the gridlines and section dividers of a colour-delimited spreadsheet.

Its executive summary, KH0083 (88 pendants, 500+ subsidiaries), exhibits SR dividers at **perfect 11-pendant intervals** (ordinals 1, 12, 23, 34, 45, 56, 67), creating a 7×10 decimal matrix. Monte Carlo validation: zero of 1,000 random colour shuffles produced any perfect periodicity for any colour ($p < 0.001$).

An eighth cluster (ordinals 78–88) falls outside the matrix and contains values 10–100× larger than the matrix cells. The ratio of the seven matrix clusters to the eighth cluster is **1.955 ≈ 2.0** — meaning Cluster 8 represents exactly **33.3%** of the total ledger. This matches the historically documented Inca Tripartite Division of Wealth: one third to the State, one third to the Sun (religion), one third to the community. KH0083 is not just a spreadsheet — it is a tax return with the state's share explicitly separated from the community's.

Furthermore, specific numerical values were repurposed as formatting. On logistical specimens like KH0504, the value 10 functions exclusively as a "tab key" or spacer, appearing at regular intervals (25% of all cords) to visually partition the data matrix rather than contribute to the sum. The clean_summation algorithm auto-detects these spacers by flagging any value appearing ≥5 times and accounting for ≥10% of pendants.

Historical "math errors" in khipu analysis — such as Karen Thompson's 4% summation gap on KH0082/KH0083 — are resolved by identifying formatting metadata embedded in the arithmetic stream. Non-standard knots (13 LL knots on KH0082, 13 EE and 3 BL on KH0083) integrate into decimal registers but do not participate in summation. These are not arithmetic errors; they are auditor stamps — non-decimal annotations tied by a reviewing official to certify the document, analogous to a modern auditor's initials on a checked ledger page. Stripping these formatting elements resolves the historical discrepancies.

3.6 Non-Standard Knot Grammar

886 non-standard knots (0.80% of 110,677 total) classify into six types: single-prime ('' — 253 occurrences, an S-type variant appearing at all register positions), SP (247, knot_value_type 2–3 at the units register), empty (152, zero-value placeholder), TF (128, all on a single khipu KH0449, a tens-register variant), EE

(75, knot_value_type 1 at units, 93%), LL (17, knot_value_type 3), and BL (14, knot_value_type 2 or 9).

All non-standard types integrate into the decimal register system rather than functioning as standalone signs. This confirms their role as positional variants and formatting annotations within the arithmetic framework — not as evidence of proto-writing or narrative encoding.

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4. The Continental Network

4.1 The Imperial Postal Route

Matched-pair analysis across 444 khipus with 10+ non-zero values identifies **3,512 pairs** sharing five or more sequential value matches at corresponding ordinal positions. Of these, **1,747 span different archaeological sites**.

Monte Carlo validation (N = 1,000): observed 3,512 pairs versus null mean 432.7 (ratio **8.1×**). Zero of 1,000 null iterations reached the observed count (**$p < 0.001$**). The matched-pair network is not an artefact of common value distributions or small number effects; it reflects genuine data synchronisation across the corpus.

The site-connection map reveals a star-topology network with three primary hubs:

Hub	Cross-Site Connections	Role
Incahuasi	467	Military garrison, logistics centre
Pachacamac	417	Religious oracle, data processing centre
Leymebamba	383	Cloud forest state archive

The top routes — Leymebamba↔Pachacamac (66 pairs, ~800 km), Incahuasi↔Pachacamac (65 pairs, ~350 km), Incahuasi↔Leymebamba (52 pairs, ~900 km) — trace the known Inca road network. The longest matched-pair connections span from Playa Miller (Arica, Chile) to the Lima region, exceeding 1,000 km of Andean terrain.

4.2 Bidirectional Data Flow

Formatting-direction analysis reveals that the network carries data in both directions:

- **1,868 Mono→Poly transitions** — census receipts flowing UP the hierarchy from provincial governor (monochrome System B original) to state archive (polychrome System A formatted copy)
- **1,286 Poly→Mono transitions** — state quotas and directives flowing DOWN from the capital (polychrome) to local administrators for execution (monochrome working copy)
- **358 Same-formatting pairs** — lateral transfers between offices at the same administrative level

This bidirectional model is consistent with the historically documented Inca mit'a (labour tax) system, in which quotas were set centrally, communicated to provincial governors, executed locally, and reported back upward.

4.3 Collection Audits: The Survival Bias Proven

Two audits of the most important published khipu collections confirm the survival bias hypothesis:

Santa Valley (Medrano/Urton). All six khipus (KH0323–KH0328) are System A or Mixed. Zero System B specimens. All cotton, all uniform S-twist (99%+), zero cluster summation. KH0326 carries canutos (postal authentication for inter-site transit) but is otherwise standard System A. Medrano's celebrated census decipherment is valid — but what he decoded was the **state copy**, not the governor's original. The census data was transcribed from a System B governance document onto System A warehouse hardware for archival storage. Two khipus (KH0325, KH0326) carry canutos — authentication seals for postal transit — confirming that these documents arrived from elsewhere.

Leymebamba Tomb. All 22 khipus from the Laguna de los Cóndores are System A. 99.2% cotton, 99.9% S-twist. Zero System B specimens. The Chachapoya elite tomb contained a state archive — formatted copies deposited with the dead as bureaucratic grave goods, not the working documents of active governance.

These audits answer the question posed in §1.2: we have been reading the receipts and wondering why they do not look like letters. The letters — System B governance originals — were carried away by runners, superseded once transcribed, and rarely preserved.

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5. Evolutionary Origin

5.1 The Wari Inheritance

Three STRUCTURE=W (wrapped-construction) khipus in the OKR represent Wari-era specimens (~600–1000 CE). Value-transition analysis demonstrates that two of three are computationally identical to Inca System B specimens across all measured dimensions:

Khipu	Era	Fibre	L-type%	Distribution	Mean	Max	Score
KH0346	Wari	Camelid	98%	Low-range	7.0	25	0.59
KH0282	Wari	Camelid+Hair	100%	Gaussian	18.2	33	0.57
KH0289	Pre-Wari?	Human hair	98%	Gaussian	26.9	46	0.50
KH0288	Inca	Camelid	100%	Gaussian	15.1	25	0.72
KH0348	Inca	Camelid	98%	Gaussian	7.4	21	0.64
KH0118	Control	Cotton	8%	Arithmetic	9,786	37,076	−0.20

The cotton control (KH0118) confirms that the similarity between Wari wrapped khipus and Inca System B is not a generic property of all khipus but a specific evolutionary lineage.

KH0289 — the "hair khipu" from the Gold Museum in Lima — extends this lineage further. Its 180 cords are 99% human hair fibre, yet its System B score (0.50 HIGH), Gaussian distribution (mean 27, max 46), 97.6% L-type knots, uniform S-twist, monochrome palette, and completely flat topology are statistically identical to the camelid System B specimens. KH0282 (Wari wrapped) contains a transition zone: camelid cords at ordinals 1–58, then human hair cords at ordinals 59–98 — two fibre traditions on a single document.

The recording *method* — L-type knots, flat structure, Gaussian headcounts, monochrome palette — is the invariant. The *material* evolved: human hair → camelid → cotton. System A (cotton, polychrome, hierarchical, S-type) appears to be an Inca-era innovation developed for empire-scale warehouse logistics. System B is the inherited tradition, potentially predating the Wari civilisation itself.

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6. The Glass Box: Methodology and Reproducibility

6.1 Rapid Hypothesis Testing via MCP

Initial exploration of the 54,403-cord corpus was conducted using a KARP (Knowledge Acquisition Research Protocol) architecture. The OKR SQLite database was loaded into an in-memory semantic graph accessible via a Model Context Protocol (MCP) server with 25 dynamic tools (classify_system, value_transitions, colour_transitions, cord_sequence, test_summation, clean_summation, matched_pairs, audit_trail, canuto_analysis, scribe_fingerprint, twist_analysis, cord_notes_search, metadata_query, and others). This architecture allowed the research team to test and falsify hypotheses in real time — over 30 findings and 9 falsifications in a single research session — at a pace impossible with traditional batch-script workflows.

6.2 Algorithmic Extraction into Standalone Scripts

No finding in this paper relies on the output of large language models. Once a pattern was identified via the MCP server, the underlying algorithm was extracted, documented, and frozen into a deterministic Python script. Four scripts reproduce the complete analytical pipeline:

- **mdp_entropy.py** — Shannon entropy computation, colour-value taxonomy, site-specific motif search. Reproduces §2.
- **mdp_classifier.py** — 6–8 dimension system classification, value distribution shape analysis, clean summation with spacer detection. Reproduces §3.
- **mdp_matched_pairs.py** — Cross-site matched-pair detection, formatting-direction analysis, Monte Carlo null-hypothesis test (N=1,000). Reproduces §4.
- **corpus_sweep.py** — Full census of all 612 khipus with system classification and value shape. Produces the N-counts cited throughout.

All scripts require only the Python 3.8+ standard library (sqlite3, math, csv, random, collections) and the publicly available OKR SQLite database. No external packages, no API keys, no proprietary tools.

6.3 Multi-AI Deliberative Design (MADD)

The research presented in this paper was conducted using the Multi-AI Deliberative Design (MADD) methodology (Sharman, 2026), in which multiple large language models contribute complementary analytical strengths through structured adversarial council sessions, with the human researcher serving as principal investigator and final arbiter.

The KARP (Knowledge Acquisition Research Protocol) platform orchestrated deliberation across four AI models. Claude (Anthropic) served as the primary computational engine: designing all algorithms, writing all Python scripts and SQL queries, building the KARP Khipu Graph MCP server with 25 analysis tools, conducting the statistical validation suite, and performing the corpus-wide classification sweep. Gemini (Google) contributed the "Spreadsheet Fallacy" conceptual framing, proposed the four-pillar paper architecture used in Paper III, generated culturally grounded hypotheses (including the Hanan/Hurin midpoint test, canutito authentication hypothesis, and Inca Tripartite Division of Wealth), and provided ethnohistorical context from Andean scholarship. Grok (xAI) contributed adversarial review and stress-testing of findings during deliberation sessions.

The human researcher (Sharman) designed and built the KARP platform, directed all research priorities, made all editorial and methodological decisions, identified the dual-system ontology from initial MCP exploration, established contact with domain experts (Thompson, Rojas Leiva), and served as the integrating intelligence across sessions. No finding in this paper was accepted without the principal investigator's independent verification.

The MADD methodology enabled a research velocity impossible through traditional workflows: 30 findings, 9 falsifications, 11 analysis scripts, and a three-layer verification suite were produced in a single extended research session. The methodology's core principle — that structured disagreement between models with different training data and reasoning approaches produces more robust findings than any single model — is itself validated by the present results. The adversarial process generated the Hanan/Hurin hypothesis (falsified), the double-entry bookkeeping hypothesis (falsified), and the calendrical sum hypothesis (falsified) alongside the 30 positive findings. The falsifications are as important as the discoveries: they demonstrate that the methodology tests ideas rigorously rather than confirming priors.

6.4 Raw SQL Verification

As a final layer, the script verify_glass_box.py generates raw SQL queries for the most critical findings — the LK knot inversion, fibre-value ratios, canutito colour independence, and others. These queries are provided in Appendix C and can be executed directly in DB Browser for SQLite against the public OKR database, requiring no programming knowledge whatsoever.

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7. Negative Findings

Nine hypotheses were formally tested and falsified during this research:

1. **Spacing as punctuation** — No significant correlation between physical cord spacing and colour-group boundaries.
2. **Hanan/Hurin midpoint** — No structural division at the document midpoint corresponding to the Inca moiety system.
3. **Cord thickness encodes domain** — No significant pattern.
4. **System C is proto-writing** — Monte Carlo bigram test rejects sequential syntax ($p > 0.05$ for both confirmed specimens).
5. **Calendrical sums** — Clean summation finds no matches for 365, 328, or 354 (solar, lunar, or festival calendars).

6. **Twist encodes value magnitude** — Only 3.7% mean difference between S-twist and Z-twist values.
7. **Value-10 spacers explain Thompson's gap** — Spacers account for negligible variance; non-standard knots are the true source.
8. **Biometric twist-angle fingerprinting** — The TWIST_ANGLE field is not recorded in the OKR.
9. **Double-entry bookkeeping via tied cords** — 28 "tied together" instances show unbalanced values (20/0, 0/0, 0/50). Tied cords are structural, not debit/credit pairs. The Inca used matrix transposition instead.

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8. Discussion

8.1 The Colour Alphabet

The three-tier entropy hierarchy has direct implications for khipu decipherment. The standard practice of analysing only primary colours (COLOR_CD_1) operates at 3.630 bits; using all 720 compound colours (FULL_COLOR) operates at 4.995 bits. The compound suffixes — barberpoles, stripes, multi-colour constructions — carry 27% of the total information. They are not decorative variation; they are data.

The site-specific colour signatures provide a framework for cross-referencing with colonial-era documentary sources. The *visitas* (colonial censuses) list tribute obligations by region; Guaman Poma's *Nueva Corónica* illustrates khipu use in specific contexts; Murra's "vertical archipelago" model documents which ethnic groups controlled which ecological zones. This paper provides the computational infrastructure — the colour vocabulary ranked by information content and mapped to geographic centres. The translation dictionary — what each colour-word *means* — is a problem for ethnohistorians equipped with colonial sources.

8.2 The Continental Database

The matched-pair network is, to our knowledge, the first computational demonstration of empire-wide data synchronisation in the pre-Columbian Americas. The three hub sites form a triangle spanning the central Peruvian coast and highlands. The chasqui postal runners were not merely carrying messages; they were maintaining a distributed database, with cryptographic authentication (canutitos) and redundant copies (matched pairs across sites) ensuring data integrity across continental distances.

The bidirectional formatting analysis adds a critical refinement: this was not a one-way reporting system. Census receipts flowed upward with formatting upgrades (monochrome → polychrome), while quota orders flowed downward with formatting downgrades (polychrome → monochrome). The Inca state operated a bidirectional API — a read-write interface between central command and provincial governance.

8.3 Why We Could Not Read Them

The survival bias is now quantified. System A (cotton, polychrome, hierarchical, arithmetic) comprises 25.2% of the corpus — but 100% of Medrano's Santa Valley, 100% of the Leymebamba tomb, and the vast majority of museum collections.

System B (camelid, monochrome, flat, demographic) comprises 4.6% — a small minority surviving in private collections and luxury containers. System C (categorical codebooks) comprises 10.3% — previously dismissed as "anomalous" or "non-conforming to base-10."

The field has been reading one type of document and assuming it represented the entire system. It is as though a future archaeologist excavated a post office, found only printed receipts, and concluded that 21st-century civilisation communicated exclusively through numbers.

8.4 Limitations

Several limitations constrain the present analysis. The OKR's FULL_COLOR field contains recording-era inconsistencies: the same physical colour may be described as KB:W by one investigator and KB-W by another, potentially inflating the unique colour count. Geographic concentration of colour signatures may partially reflect differential access to dye materials. The matched-pair algorithm uses a simplified sequential comparison that does not account for ordinal shifts. The System B sample (28 specimens) and System C sample (63) remain small; further fieldwork targeting camelid-fibre collections would strengthen the statistical foundation. The hair-fibre evolutionary claim (§5.1) rests on a single specimen (KH0289) and requires additional dating and provenance data.

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9. Conclusion

The Inca khipu is not a spreadsheet. It is a multi-system recording technology operating through three administrative traditions (logistical, demographic, and categorical), two complementary formatting channels (colour and twist), and a continental data network validated at $p < 0.001$. The standard practice of collapsing compound colours discards 27% of the colour signal. The survival bias of the archaeological record has preserved the arithmetic receiver-end copies while the demographic sender-end originals have largely perished.

We do not claim to have decoded the khipu. We claim to have identified the alphabet, measured the information content of each letter, mapped the geographic distribution of each word, proved that the postal network carried synchronised data across 1,000 kilometres, and explained why 100 years of research produced the impression that these documents were "just arithmetic."

The decipherment — the translation of colour-words into commodities, lineages, and administrative categories — awaits ethnohistorians equipped with colonial sources and the computational infrastructure this paper provides.

Appendix A: Comprehensive Findings Inventory

A.1 Tier 1 Findings (Paradigm-Level)

F1. Dual-System Ontology. The Inca Empire operated two parallel administrative systems on different media. System A (Cotton/Logistical): polychrome, hierarchical (4–5 levels), S-type dominant (68–90%), arithmetic values, decimal groupings. Found at Incahuasi, Pachacamac. System B (Camelid/Demographic): monochrome, flat (zero subsidiaries), 100% L-type, Gaussian values (1–33), canuto-authenticated. Found across 9 provenances, 14 investigators. Corpus-wide classification: 154 System A (25.2%), 28 System B (4.6%), 367 Mixed (60.0%).

F2. Wari → System B Evolutionary Lineage. Three wrapped-construction khipus (STRUCTURE=W) match Inca System B across all dimensions. KH0346 (Wari, camelid, 98% L-type, score 0.59), KH0282 (Wari, camelid+hair, 100% L-type, 0.57). Cotton control KH0118 (8% L-type, −0.20) confirms specificity. System A is the Inca innovation; System B is inherited Wari technology.

F3. Redundant Channel Principle. Polychrome khipus use colour for formatting and twist is uniform. Monochrome khipus use twist for formatting (Z-density phases 8%→83% on KH0288) and colour is uniform. Each system uses one channel for structure and the complementary channel for data, ensuring no interference.

F4. Forbidden Colour Transitions. B→NB = 0 observed versus 38 expected. Strict colour-sorting grammar governs cord sequencing (Paper III §4.1).

F5. Two-Stage Administrative Pipeline. Same commodity (peanuts) recorded as NB locally, W in state copies. Formatting upgrades (mono→poly) = receipts flowing UP. Formatting downgrades (poly→mono) = quotas flowing DOWN. KH0491→KH0503 canonical example.

A.2 Tier 2 Findings (Major)

F6. Imperial Colour Punctuation. KH0082 uses 5 colour classes as zero-value structural markers: SR (16 cords, 100% zero, gap 105), KB (32, 97%), HB (8, 100%), FB (13, 100%), BS (25, 92%). KH0083 SR dividers at perfect 11-pendant intervals = 7×10 decimal matrix. $p < 0.001$ (0/1,000 null).

F7. Corpus-Wide Colour Matrices. 95 khipus scanned. KH0468 (Ica-Pisco): 87 HB markers in 7 blocks, 89% zero — largest matrix. All matrix khipus are System A at major administrative centres.

F8. Audit Trail — System B Immutability. Only 7 cut cords in 54,403 (0.013%). 6/7 on KH0321 (header redaction, System A). Insertions also exclusive to System A/Mixed. Zero physical edits on any System B specimen.

F9. System C = Categorical Codebook. 63 Mixed-classification khipus exhibit Zipfian distributions. Monte Carlo bigram test rejects syntax (KH0100: 4.8% vs 4.0%; KH0108: 8.7% vs 7.1%). Shared 11-value vocabulary: 9,10,11,12,14,17,18,19,20,21,22 covering ~47% of each specimen. Not proto-writing — unordered categorical labels.

F10. Cross-System Bridge. 11 values shared across Systems A, B, and Mixed. Two exact 3-value sequences match: (11,11,11) and (14,11,20). 94% of System B values appear on Pachacamac.

F11. System B Expanded to 28 Specimens. Corpus-wide sweep identifies 28 System B khipus (up from 8 manual), with 7 HIGH confidence: KH0288 (0.72), KH0280 (0.67), KH0348 (0.64), KH0346 (0.59), KH0282 (0.57), KH0384 (0.57), KH0289 (0.50, human hair).

F12. S/Z Twist Partition (KH0288). Z-density phases from 8%→83%. Z capped at value 20. S-twist cords (6/115) are rare dividers. Validates Redundant Channel

Principle on System B.

F13. First-Cord Header = System Identifier. System A opens high (196–1,620). System B opens low (5–19).

F14. Institutional Manufacturing Fingerprint. System B cord-length CV: KH0288 = 11.1% (pre-planned, uniform). System A CV: KH0491 = 30.4%, KH0503 = 32.9% (utilitarian, variable). Incahuasi pair within 5% = same workshop.

F15–F17. Paper III Validated Findings. Fibre redefines domains (cotton mean=294 vs camelid mean=7, 41×). LK anomaly (79.5% L-type, χ^2=2797). 648 cross-colour matched pairs. Incahuasi cluster summation ratio=1.00. Divider specialist spectrum (LC=87.6%). NB 98× enriched at Incahuasi. Canutito independence (98.6%).

F18. Thompson Clean Math. KH0082 gap caused by 13 LL non-standard knots (kvt=3, turns=0), NOT value-10 spacers (only 24 spacers). KH0083 has 13 EE + 3 BL + 4 empty. Non-standard knots are auditor stamps, not arithmetic errors. Baseline KH0082=12,919 (matches Thompson 12,915 ±4).

F19. Double-Entry Bookkeeping Falsified. 28 "tied together" instances corpus-wide. KH0082 pair (ord 321–322): both value 0. KH0508 pairs: 20/0, 0/0, 0/50. NOT balanced. Tied cords are structural. Inca used matrix transposition instead.

F20. Imperial Data Network. 3,512 matched pairs (1,747 cross-site). Monte Carlo: observed/null ratio = 8.1×, $p < 0.001$. Top routes: Leymebamba↔Pachacamac (66 pairs, ~800 km), Incahuasi↔Pachacamac (65, ~350 km), Incahuasi↔Leymebamba (52, ~900 km). Three hubs: Incahuasi (467 connections), Pachacamac (417), Leymebamba (383). Bidirectional: 1,868 mono→poly (receipts UP), 1,286 poly→mono (quotas DOWN).

F21. Clindaniel Non-Standard Knot Grammar. 886 non-standard knots (0.80% of 110,677). Six types: ’’ (253, S-variant), SP (247, kvt=2–3 units), empty (152, kvt=0), TF (128, all KH0449 tens-register), EE (75, kvt=1 units 93%), LL (17, kvt=3), BL (14, kvt=2/9). All integrate INTO decimal register — not standalone signs.

F22. Shannon Entropy Colour Dictionary. 39,172 pendants, 720 unique FULL_COLORs. Corpus entropy: 4.995 bits (52.6% of max 9.492). Information lost by COLOR_CD_1 collapse: 1.366 bits (27%). Three-tier hierarchy: Vowels (<4 bits): W(1.89), AB(2.88), MB(3.35). Semi-vowels (4–7): B, YB, KB, LB, AB:W, NB, HB, LK. Consonants (7–10): RL, BG, GL(9.59). Rare consonants (10+): SR(11.01). Site signatures: LK=Pisco(98%), NB=Incahuasi(100%), G=Nazca(95%), RL=Santa(100%).

F23. KH0279 Multi-Section Matrix. 238-cord document, 14+ FULL_COLORs including compounds. KB compound delimiters at ord 81 (KB:GG) and 166 (KB:W) split document into 3 macro-sections. GL section (ord 204–209) carries highest values (515–624) on rarest colour.

F24. Colour Functional Taxonomy. Three-layer encoding: Tier 1 Formatting (SR=100% zero, KB:GG/LC/SY=100% zero) → Tier 2 Category labels (AB/MB/W vowels, any value) → Tier 3 Domain identifiers (LK=people mean 21, NB=goods mean 516). KB function tracks System A/B boundary (formatting at coast, data at provinces).

F25. Santa Valley Audit (Medrano Gold Standard). All 6 khipus (KH0323–KH0328) are System A or Mixed. Zero System B. Uniform S-twist (99%+). 0% cluster summation. Medrano decoded the STATE COPY, not the governor's original. Canutos on KH0325/KH0326 = postal authentication.

F26. KH0328 Oscillation Grammar. Perfect 5-colour cyclic pattern: LG(6)→RL(6)→AB(6)→W(6)→RB(6), 60 cords. Values only 0–9 (mean=2.9, max=9). NOT a ledger — a TEMPLATE or CODEBOOK. RL section confirms Santa-specific signature from F22.

F27. KH0585 The Alien Khipu. Mixed(−0.13). Cotton + monochrome + hierarchical (depth 3) + canutos = traits from ALL three systems. Pendant 1 has 8 subsidiaries (values 60,110,110,0,10,0,0,30 = total 320). L-knots with value=0 (physical knots without numerical value). Possibly non-decimal.

F28. Leymebamba Tomb = State Archive. 22 khipus from Laguna de los Condores. 99.2% cotton, 99.9% S-twist. ALL System A. Zero System B. The Chachapoya elite tomb contained state archive copies, not governor originals. Confirms pipeline: what survives in tombs is the RECEIVER end.

F29. Human Hair Fibre = Pre-Wari System B Material. KH0289 (Gold Museum Lima, "hair khipu"): 100% human hair, 180 cords, System B 0.50 HIGH, Gaussian (mean=27, max=46), 97.6% L-type, uniform S-twist, monochrome, completely flat. KH0282 (Wari wrapped) has 39 hair cords at ordinals 59–98 = transition zone. Material evolution: Hair → Camelid → Cotton. Recording METHOD is the constant; material evolved.

F30. Corpus-Wide Census. 612 khipus classified. System A=154 (25.2%), System B=28 (4.6%), System C=63 (10.3%, Mixed+Zipfian), Mixed=367 (60.0%). Value shapes: Arithmetic 260 (42.5%), Zipfian 175 (28.6%), Too-few 99 (16.2%), Low-range 33 (5.4%), Binary 33 (5.4%), Gaussian 11 (1.8%). Gaussian strongly predicts System B (6/11). Systems B and C are near-disjoint (1 Zipfian in System B).

A.3 Falsifications

#	Hypothesis	Method	Result
1	Spacing as punctuation	Corpus analysis	No signal (46.1% vs 45.1% shift rate)
2	Hanan/Hurin midpoint division	Corpus analysis	Identical left/right distributions
3	Cord thickness encodes domain	Corpus analysis	No significant pattern
4	System C = proto-writing	Monte Carlo bigram (N=1,000)	Not significant (4.8% vs 4.0%)
5	Calendrical sums (365/328/354)	Clean summation	Zero matches
6	Twist encodes value magnitude	Twist analysis	Only 3.7% mean difference
7	Value-10 spacers explain Thompson gap	thompson_clean_math.py	Spacers negligible; non-standard knots are cause

#	Hypothesis	Method	Result
8	Biometric twist-angle fingerprint	scribe_fingerprint	TWIST_ANGLE field not recorded in OKR
9	Double-entry bookkeeping via tied cords	cord_notes_search	Values unbalanced (20/0, 0/50); structural, not debit/credit

Appendix B: Document Type Taxonomy

The following table categorises the novel khipu document classes computationally identified in this study, explicitly listing their Open Khipu Repository (OKR) identifiers for independent visual and physical verification.

OKR ID	Document Type / Nickname	System	Structural Fingerprint & Significance
KH0348	The Cedar Box Treasure	System B	Purest LK (Black) anomaly. 58% LK, camelid fibre, 98% L-type knots, canutito seals, flat topology. Stored in a luxury container; elite governor's record.
KH0288	The Gaussian Census	System B	Highest System B score (0.72). 98% DB (Deep Brown), 100% L-type knots, flat topology. Values form a perfect Gaussian bell-curve (mean=15, max=25). S/Z twist provides binary formatting.
KH0346	Wari Wrapped Ancestor	System B	STRUCTURE=W (Wrapped cords). 98% L-type knots, camelid fibre. Computationally proves System B is a continuation of Middle Horizon Wari administrative technology.
KH0282	Wari Wrapped (Hair Transition)	System B	STRUCTURE=W, camelid fibre with 39 human hair cords at ordinals 59–98. Transition zone between pre-Wari hair tradition and Wari camelid tradition on a single document.
KH0289	The Hair Khipu	System B	Gold Museum Lima. 100% human hair fibre, 180 cords, Gaussian (mean=27, max=46), 97.6% L-type. Purest data-only document in corpus: zero formatting, zero structure, just 180 headcounts. Extends System B lineage to pre-Wari origin.

OKR ID	Document Type / Nickname	System	Structural Fingerprint & Significance
KH0100	The 19-Knot Enigma	System C	Uses 19 different knot types. Dismissed by early researchers as having “no numerical interpretation.” Computationally proven to be a Zipfian (power-law) categorical vocabulary. Munich.
KH0108	The Pachacamac Anomaly	System C	23 distinct knot clusters on System A hardware, but values follow a Zipfian distribution. Exhibits forbidden value transitions (21–50 → 1–5 = 0 occurrences).
KH0082	Thompson Local Ledger	System A	Largest khipu in corpus (1,831 cords). 5 colour classes as zero-value structural markers (SR/KB/HB/FB/BS). The definitive colour-delimited spreadsheet.
KH0083	Thompson Executive Summary	System A	88 pendants, 500+ subsidiaries. SR dividers at perfect 11-pendant intervals = 7×10 decimal matrix. Cluster 8 = 33% of total (Tripartite Tax).
KH0468	The Largest Colour Matrix	System A	87 HB markers in 7 blocks, 89% zero. Ica-Pisco. The largest discovered colour-delimited matrix structure.
KH0279	The Multi-Section Matrix	Mixed	238 cords, 14+ compound FULL_COLORs. KB compound delimiters split document into 3 macro-sections. GL section carries highest values on rarest colour.
KH0622	The Blank Clipboard	—	10 cords, 100% LG (Light Green), 100% L-type knots. All values = 0. Found in Sector B (elite offices) of Incahuasi. An unused, mass-manufactured inspection template.
KH0623	The Completed Checklist	System A	9 cords, 100% LG, 100% E-type (Figure-8) knots. All values = 1. The completed version of the KH0622 template, tracking presence/absence (binary) rather than volume.
KH0589	The Ultimate Blank	—	33 cords, zero knots. FR-dominant. Held at the Dallas Museum. Represents the pre-data manufacturing stage of Inca administration.

OKR ID	Document Type / Nickname	System	Structural Fingerprint & Significance
KH0328	The Oscillation Grammar	System A	60 cords, perfectly balanced 5-colour cycle repeating twice. Values constrained to 0–9. The 6th cord of each group is always a zero-terminator. A lookup table or seasonal calendar.
KH0587	The Edited Document	System A	Two cords physically sewn into the primary frame between pendants 110–111. Proves System A documents were “living,” editable logistics ledgers (unlike immutable System B).
KH0449	The Pure TF Khipu	Mixed	128 of 129 knots are non-standard TF type at tens register. Ica-Pisco. A document written entirely in a variant knot notation.
KH0585	The Alien Khipu	Mixed	Cotton + monochrome + hierarchical (depth 3) + canutos = traits from ALL three systems. L-knots with value=0. Possibly non-decimal counting system.
KH0321	The Redacted Document	System A	6 cut cords at ordinals 1–6 (header removed). The only khipu with deliberate physical censorship.
KH0491/KH0503	The Ají Pipeline Pair	System A	KH0491 = monochrome AB draft (raw warehouse data). KH0503 = polychrome W+AB formatted copy (state archive version). Same chili pepper data, different administrative tiers.

Appendix C: Glass Box SQL Queries

The following 8 raw SQL queries can be executed directly in DB Browser for SQLite against the publicly available OKR database (DOI: 10.5281/zenodo.18025748). No programming knowledge is required. Each query independently verifies a key finding reported in this paper.

Q1 — LK Colour Anomaly (F15). L-type percentage by colour. Expected: LK ~79.5% L-type (highest of any colour).

Q2 — Fibre Redefines Value Domains (F15). Mean pendant value by fibre type. Expected: CN (cotton) mean ~294, CL (camelid) mean ~7, ratio ~41×.

Q3 — Canutito Independence (F15). Colour divergence between subsidiary and parent cords. Expected: ~98.6% different colours.

Q4 — Wari Wrapped Khipus = System B (F2). Knot profiles of all STRUCTURE=W specimens. Expected: 3 khipus, 2/3 with >95% L-type.

Q5 — Zero-Value KB Structural Cords (F6). KB, SR, HB, FB, BS zero-value rates on KH0082. Expected: SR 100%, KB ~97%, HB 100%.

Q6 — SR Dividers at Perfect Intervals (F6). SR cord ordinal positions on KH0083. Expected: 1, 12, 23, 34, 45, 56, 67 (period = 11).

Q7 — Cut Cords / Audit Trail (F8). All cords with TERMINATION='C'. Expected: 7 rows, 6 on KH0321 (ord 1–6), 1 on KH0526.

Q8 — Non-Standard Knot Census (F21). All knot TYPE_CODE counts. Expected: S ~75K, L ~26K, E ~8K, non-standard ~886 (0.80%).

Full SQL source code is provided in the supplementary file `verify_findings.sql` and in the project repository at github.com/souldriver007/mdp-ancient-scripts.

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